

SOME POPULAR MCUs AVAILABLE IN THE MARKET

Product/Family	Company	Features	Applications	Architecture
PIC24FJ 1024GA610/ GB610 family 	Microchip	- PIC24F 16-bit MCU featuring large dual-partition flash with live update capability - Family includes up to 1024kB flash, 32kB RAM, USB and advanced peripherals. A combination of features makes the part ideally suited for always-on applications and applications requiring high memory	—	16-bit
Apollo ultra-low-power MCU 	Ambiq Micro	- Features active mode current in the order of 34µA/MHz when running from flash and sleep mode current less than 150nA - Available with a range of memory options from 64kB to 512kB of embedded flash memory - Includes 10-bit ADC - Offered in both BGA and WLCSP package options	- Wearable electronics - IoT devices - Wireless sensors - Consumer medical devices - Consumer electronics - Smartwatches - Smartmeters - Battery-powered applications	32-bit
Arduino Yun 	Arduino	- Based on ATmega32u4 and Atheros AR9331 - The board has built-in Ethernet and Wi-Fi support, USB-A port, micro-SD card slot, 20 digital input/output (I/O) pins (of which seven can be used as PWM outputs and 12 as analogue inputs), 16MHz crystal oscillator, micro-USB connection, ICSP header and three reset buttons - Onboard Linux distribution	—	8-bit
EFM8LB1 [EFM8 Laser Bee 8-bit (MCUs) family] 	Silicon Labs	- Pipelined 8-bit 8051 MCU core with 72MHz operating frequency - Up to 29 multifunction I/O pins - One 14-bit, 900 ksp/s ADC - Up to four 12-bit DACs with synchronisation and PWM capabilities - Two low-current analogue comparators with built-in reference DACs - Internal calibrated temperature sensor ($\pm 3^{\circ}\text{C}$) - Internal 72MHz and 24.5MHz oscillators accurate to $\pm 2\%$ - Four channels of configurable logic 6-channel PWM/PCA - Six 16-bit general-purpose timers	- Optical network modules - Precision instrumentation - Industrial control and automation - Smartsensors	8-bit
MSP430FR2633 	Texas Instruments	- Lets designers build HMI with capacitive buttons, sliders, wheels or proximity (BSWP) sensors - Features non-volatile memory and can support 16-button self-capacitance as well as 64-button mutual-capacitance modes - Wake-on-touch hardware on chip can continuously scan up to four electrodes while CPU is shut down	- Thermostats - White goods - Electronic access control - Small appliances - Lighting control - Personal electronics - Electronic door-locks	16-bit RISC
SMART SAM E70 	Atmel	- Has up to 2048kB of flash and 384kB of configurable SRAM, and offers up to 114 programmable I/O pins as well as an ultra-low-power real-time clock and 32-bit ultra-low-power real-time timer - Dual 12-bit analogue-to-digital converters with analogue front-end offering offset and gain error correction - Twelve general-purpose 16-bit timers	- IoT or smartenergy gateways - Industrial automation - Building control applications	32-bit
TMPM066FWUG, TMPM067FWUG, TMPM068FWXBG (under development) 	Toshiba	- ARM Cortex-M0 core - Maximum frequency: 24MHz - Built-in USB device controller - Small pin packages - Multiple serial interface function including SPI, I2C (fast mode plus) and others - Bit-banding	- Tablets - PCs - Digital equipment - Industrial controllers - Office automation equipment - Sensor hubs	—

MCUs in general are an integral part of IoT use cases. While the IoT encompasses a lot of applications (both battery-powered and line-powered), MCUs in the future

would see integrated connectivity (both wired and wireless)."

Rise of Type-C

Navaneethan Sundaramoorthy, co-

founder and CMO, Uncanny Vision Solutions, explains that the most exciting feature for him in the last 18 months has been the introduction of universal serial bus (USB) Type-C in-